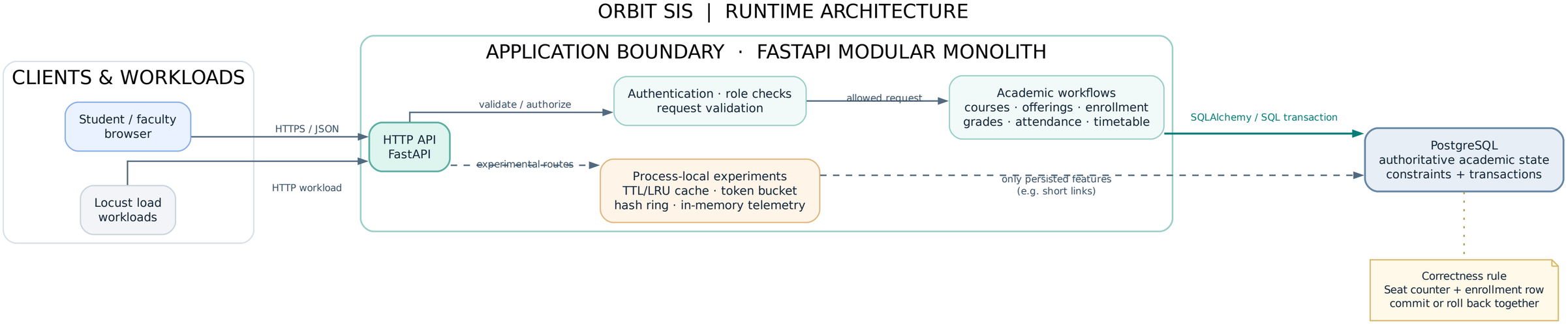
ORBIT SIS | RUNTIME ARCHITECTURE
CLIENTS & WORKLOADS
Student / faculty browser
Locust load workloads
HTTPS / JSON
HTTP workload
APPLICATION BOUNDARY · FASTAPI MODULAR MONOLITH
HTTP API FastAPI
validate / authorize
Authentication · role checks request validation
allowed request
Academic workflows courses · offerings · enrollment grades · attendance · timetable
experimental routes
Process-local experiments TTL/LRU cache · token bucket hash ring · in-memory telemetry
only persisted features (e.g. short links)
SQLAlchemy / SQL transaction
PostgreSQL authoritative academic state constraints + transactions
Correctness rule
Seat counter + enrollment row commit or roll back together